

COURSE 3 - DATA ENGINEERING 2026

C3-082 - Cloud & Microsoft Fabric Data Engineering Protected Review

Open only after matching Pxx attempt. Diagnose and transfer; do not copy as a finished solution.

Artifact	C3-082
Status	2026 - LEARNER EDITION
Phase	C3-07 Cloud & Microsoft Fabric Data Engineering 15 lessons 50-70 focused hours
Prerequisite	C3-06 Gate PASS; C2B cloud foundations are recalled, while Fabric implementation is taught from first principles.
Current lane	Microsoft Fabric docs refreshed 2026-09-11 trial 60 days F4/F64 up to 1 TB OneLake
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided Fabric action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> security/production reasoning -> evidence. A portal screenshot is not mastery unless the underlying data, identity, lifecycle and recovery claim is independently verified.

Protected-use rule

Compare one attempted task, identify the exact weak competency, close this file, then solve a changed case. Failed attempts stay append-only.

R01 - Review for P01

What strong evidence shows

Boundaries separate storage, compute, connectivity/control and identity; the service choice follows the requirement.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If your diagram equates workspace with storage or says SaaS removes IAM/cost/recovery duties, redraw the boundaries.

Changed retry

Move the source behind a private/on-prem network and add the required connectivity decision.

Close this review. Complete the changed retry separately and preserve the original attempt.

R02 - Review for P02

What strong evidence shows

Least-privilege matrix separates workspace/item action from OneLake data access and proves a negative test as the consumer.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If you solve data access by broad Contributor membership, separate item management from data-reading needs.

Changed retry

Change a consumer from analyst to ingestion service identity and redesign permissions.

Close this review. Complete the changed retry separately and preserve the original attempt.

R03 - Review for P03

What strong evidence shows

Shortcut/mirroring/movement/native choices follow ownership, freshness, transformation and recovery needs.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If every source is copied, ask whether a no-copy shortcut meets freshness/governance. If every source is a shortcut, ask where transformation/ownership belongs.

Changed retry

Change a source from static files to operational database CDC and re-evaluate.

Close this review. Complete the changed retry separately and preserve the original attempt.

R04 - Review for P04

What strong evidence shows

Raw Files and trusted Delta Tables have explicit contracts, grain, validation and safe correction behavior.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If "uploaded successfully" is your only evidence, add row/key/schema/business-total controls.

Changed retry

Add one corrected order and prove an idempotent Silver result.

Close this review. Complete the changed retry separately and preserve the original attempt.

R05 - Review for P05

What strong evidence shows

Movement/orchestration/visual transform/code-heavy transform are not conflated; Copy Job is considered for simple movement.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If you choose Dataflow merely because it is visual, restate transformation complexity/scale/lifecycle needs.

Changed retry

Change a simple copy to a schema-heavy transform and re-evaluate.

Close this review. Complete the changed retry separately and preserve the original attempt.

R06 - Review for P06

What strong evidence shows

Connection, parameter, gateway, credential and network path are separately reasoned and tested.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If every source gets a gateway, separate authentication from network reachability.

Changed retry

Move a cloud database behind a private network and re-evaluate.

Close this review. Complete the changed retry separately and preserve the original attempt.

R07 - Review for P07

What strong evidence shows

Notebook code is explicit, reusable and validated; a successful cell alone is not sufficient evidence.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If notebook success depends on hidden session state, restart/re-run from clean context and externalize parameters.

Changed retry

Run the same logic against incremental input or a different workspace item.

Close this review. Complete the changed retry separately and preserve the original attempt.

R08 - Review for P08

What strong evidence shows

Spark job/environment/lakehouse references are controlled and remapped across workspaces with post-deploy smoke evidence.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If deployment assumes source workspace IDs work everywhere, add explicit remapping/validation.

Changed retry

Promote Dev -> Test with a different Lakehouse/environment name.

Close this review. Complete the changed retry separately and preserve the original attempt.

R09 - Review for P09

What strong evidence shows

Warehouse serving layer has explicit relational grain, objects, permissions and reconciliation.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If you copy raw data just to get SQL access, ask whether Lakehouse SQL endpoint is enough.

Changed retry

Add a second serving mart requiring relational constraints/schema ownership and revisit Warehouse choice.

Close this review. Complete the changed retry separately and preserve the original attempt.

R10 - Review for P10

What strong evidence shows

Eventstream/Eventhouse/KQL path matches event-time use cases and is not confused with relational serving.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If you put all historical relational serving into Eventhouse, restate query pattern, latency and model needs.

Changed retry

Add a near-real-time alert/Activator need and revise the path.

Close this review. Complete the changed retry separately and preserve the original attempt.

R11 - Review for P11

What strong evidence shows

Incremental state advances safely; update/insert/delete semantics and rerun convergence are proven.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If you move the watermark before validating the target, repair commit ordering.

Changed retry

Add a late correction whose updated_at is newer but business date older.

Close this review. Complete the changed retry separately and preserve the original attempt.

R12 - Review for P12

What strong evidence shows

Git/deployment controls definitions/metadata while OneLake data is validated separately; preview-sensitive references are called out.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If your plan says "Git backs up the data," separate item definition/metadata from OneLake table/file data.

Changed retry

Introduce a Test workspace with different connection/environment IDs and revise promotion.

Close this review. Complete the changed retry separately and preserve the original attempt.

R13 - Review for P13

What strong evidence shows

Monitoring evidence connects run/platform/business signals to detect/diagnose/recover/validate actions.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If the first response to slowness is "increase capacity," prove whether source/sink/query design or concurrency is the actual bottleneck.

Changed retry

Change an incident from compute saturation to source throttling and revise response.

Close this review. Complete the changed retry separately and preserve the original attempt.

R14 - Review for P14

What strong evidence shows

Security proof uses the intended Viewer/consumer identity and accounts for other grant paths; masking is not authorization.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If you test only as workspace Admin, your access proof is invalid. Use a consumer identity/role plan and negative checks.

Changed retry

Add a contractor who may read East-region orders without customer email.

Close this review. Complete the changed retry separately and preserve the original attempt.

R15 - Review for P15

What strong evidence shows

Capacity plan uses observed workload/CU evidence and separates certification mapping from engineering mastery.

Check your attempt

- Correct source/target/resource boundary
- Acting identity and permission path explicit
- Independent correctness/security control present
- Live platform claim supported by observed evidence when claimed
- Failure/recovery and changed transfer are reproducible

Targeted repair

If your DP-700 map lists topics you have never implemented/evidenced, mark them as study gaps, not mastery.

Changed retry

Cut available capacity in half and reschedule workloads while preserving SLA.

Close this review. Complete the changed retry separately and preserve the original attempt.